

Research

Design and circuit applications of PTnMOSFET devices from planar to gate-all-around structures

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Abstract

In this work, we propose a punch-through nMOSFET (PTnMOSFET) as a practical alternative to pMOSFETs by adjusting the source/drain doping concentrations. This PTnMOSFET design simplifies device fabrication by eliminating the need for sizing adjustments or stress engineering techniques while effectively addressing current mismatch issues between pMOSFETs and nMOSFETs. Simulations of Planar, Fin Field-Effect Transistor (FinFET), and Gate-All-Around (GAA) PTnMOSFETs reveal that the GAA PTnMOSFET achieves a wide design window, an average subthreshold swing (SS_{avg}) of 74 mV/dec, and an $I_{ON(PTnMOSFET)}/I_{ON(nMOSFET)}$ ratio of 0.89. Gate lengths (L_G) of 30 nm and 14 nm were selected to align with current and emerging semiconductor technology nodes, ensuring relevance to state-of-the-art devices. In circuit applications, the GAA single-carrier (SC) circuit exhibits significant improvements over the conventional (Conv.) GAA circuit (with $W_p/W_n = 1$). The GAA SC circuit demonstrates a 14.9% shorter delay time, a 5.5% higher voltage gain, and more symmetric signal noise margins (NM), with 117 mV for signal noise margin high (NM_H) and 118 mV for signal noise margin low (NM_L). Benchmarking results further suggest that under the same supply voltage (V_{DD}) and number of transistors, our GAA SC Static Random Access Memory (SRAM) achieves a read failure voltage of 0.145 V and a static signal noise margin of 56 mV in read mode, surpassing state-of-the-art studies on stacked nanosheet and cFET counterparts. While these results highlight the strengths of the proposed design, further experimental validation under diverse operating conditions is essential to fully assess its robustness and practical applicability.

Keywords Punch through nMOSFET (PTnMOSFET) · Hole mobility · Gate-all-around (GAA) · Average subthreshold swing (SS_{avg}) · Read static noise margin (SNM_{Read}) · SRAM · Single-carrier (SC)

1 Introduction

The continuous evolution of MOSFETs has propelled the semiconductor industry into the angstrom era, characterized by fin width shrinking to as low as 2 nm. This advancement aligns with the shift from Fin Field-Effect Transistor (FinFET) technology to more advanced gate-all-around (GAA) CFET architectures, aimed at maximizing current-carrying capability through the multi-layer stacking of nanosheets to improve overall device drive current [1–8]. However, this multilayer approach introduces significant fabrication challenges, including precise etching between stacked layers and the management of parasitic effects that compromise device yield, increase variability, and raise production costs.

Despite these benefits, GAA structures face scaling limitations when further reduced below 5 nm due to increased reliability concerns and thermal effects in ultra-short channels. The challenges posed by short-channel effects (SCEs) become more pronounced as scaling continues, affecting both device performance and manufacturability. Solutions such as pocket doping,

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as explored in planar silicon on insulator (SOI) junctionless transistors, have shown improvements in subthreshold swing (SS) and I_{ON}/I_{OFF} ratios, contributing to enhanced device behavior at smaller scales [9]. FinFET based SRAMs have demonstrated significant advancements in stability and I_{ON}/I_{OFF} ratios, making them suitable for low-power applications in compact devices [10]. However, these strategies may not fully address the inherent limitations faced by ultra-scaled devices [11, 12].

Amidst these scaling challenges, the emergence of punch-through nMOSFET (PTnMOSFET) and single-carrier Complementary Metal Oxide Semiconductor (SC CMOS) architectures presents a promising alternative. PTnMOSFET technology leverages a non-destructive punch-through current mechanism, where the extent of source/drain depletion is influenced by gate bias rather than traditional saturation or linear current modes. This allows for superior control over subthreshold leakage and improved resistance to SCEs, as demonstrated in prior research on planar PTnMOSFETs [13–15]. The punch-through behavior becomes particularly beneficial as device gate lengths shrink, with studies indicating enhanced reliability and noise immunity in planar SRAM configurations compared to their conventional counterparts.

This study focuses on PTnMOSFETs with gate lengths of 30 nm and 14 nm to reflect both current planar technology and future GAAFET nodes, ensuring relevance across generations. The SC CMOS design maximizes the use of high-mobility materials, such as silicon–germanium (SiGe) and certain III-V compounds, which facilitate simpler manufacturing processes by reducing dependency on doping and mitigating the effects of random doping fluctuation (RDF). This approach ensures minimal thermal budget requirements and robust integration with current CMOS fabrication processes. SC CMOS designs utilize lightly doped or near-intrinsic source/drain regions, resulting in smoother energy profiles across the channel and promoting effective punch-through characteristics, supporting consistent low-power operation while maintaining high performance. Additionally, exploring alternative materials, such as transition metal dichalcogenides (TMDs) like MoS₂, could significantly enhance PTnMOSFET performance due to their superior carrier mobility and reduced effective mass. Integrating these materials into GAAFET configurations promises improved subthreshold characteristics and higher current drives, particularly advantageous for ultra-scaled devices.

The motivation behind this study stems from the opportunity to expand the research on PTnMOSFETs beyond planar implementations to include advanced 3D FinFET and GAA configurations. Previous studies have highlighted the superior resistance to noise and improved subthreshold characteristics of planar PTnMOSFETs [13]. However, applying these findings to vertically stacked nanosheet structures, such as GAAFETs, remains a relatively unexplored domain. The GAAFET architecture, with its superior electrostatic gate control and lower susceptibility to SCEs, presents an ideal platform for harnessing the full potential of PTnMOSFETs in highly scaled nodes.

The primary objective of this paper is to thoroughly analyze PTnMOSFET performance across different structures (planar, FinFET, and GAAFET) and to showcase their comparative benefits in terms of SS , Drain Induced Barrier Lowering (DIBL), and design windows. The GAAFET PTnMOSFET, in particular, is shown to provide a broader design window and the lowest average subthreshold swing (SS_{avg}), which are crucial for maintaining device efficacy as scaling progresses.

Additionally, this research aims to integrate PTnMOSFETs into circuit-level designs, including CMOS and SRAM circuits, to evaluate their performance in real-world applications. The circuit-level analysis includes assessing delay times, noise margins, and voltage gain, positioning SC CMOS and SC SRAM configurations as advantageous over traditional designs with W_p/W_n ratios of 1 and 3. This work emphasizes the strengths of SC CMOS in achieving low-power, high-performance outputs that are essential for next-generation applications such as IoT, 6G communications, and AI systems. The study's benchmarking highlights that the GAAFET SC SRAM exhibits the lowest read failure voltage and the highest SNM_{Read} among evaluated designs, indicating its strong potential for future low-voltage and high-performance applications.

In summary, this research not only reinforces the feasibility of PTnMOSFETs within conventional CMOS process flows but also extends the potential of SC CMOS technology as an efficient solution for continued scaling. The PTnMOSFET approach, with minimal process modifications and cost-effective integration, presents a forward-thinking response to the scaling challenges faced in semiconductor advancements.

2 Structural design and simulation method

Figure 1 shows the planar structure of conventional CMOS circuit and our proposed SC CMOS circuit. Figures 2 and 3 illustrate the structures of the FinFET, and GAAFET SC CMOS circuit, with cross-sectional views along the A–A' and B–B' planes. Their respective device parameters are listed in Tables 1, 2, and 3. The source and drain doping concentrations ($2 \times 10^{17} \text{ cm}^{-3}$) are intentionally set very close to the channel concentration ($1 \times 10^{17} \text{ cm}^{-3}$) to enhance the depletion regions formed within the channel by the source and drain, facilitating electron punch-through from the source to the drain.

Fig. 1 Schematic cross-sections of (a) a conventional Planar CMOS and (b) the proposed SC CMOS

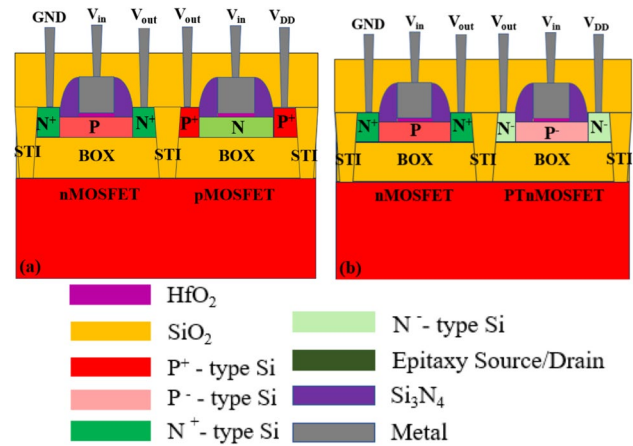
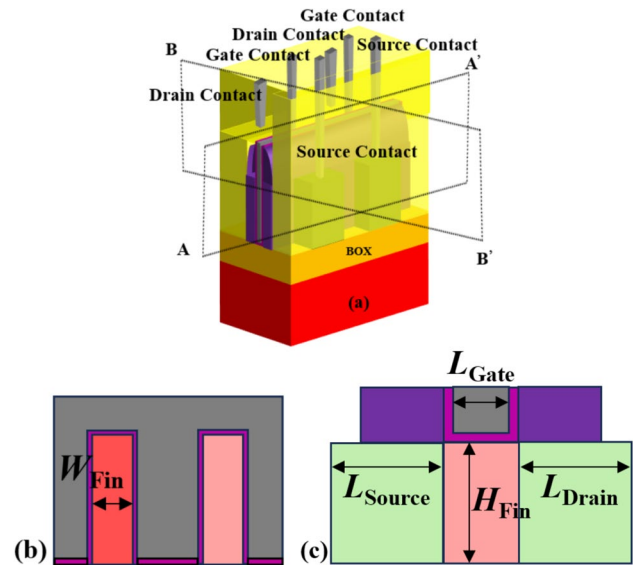


Fig. 2 (a) FinFET SC CMOS Proposed and its cross-sectional views along (b) A-A' plane and (c) B-B' plane, respectively



To ensure a comprehensive evaluation of PTnMOSFET performance, gate lengths (L_G) of 30 nm and 14 nm were selected for the simulations. The initial analysis utilized a 30 nm gate length to facilitate a fair comparison between planar PTnMOSFET and its evolved structures (e.g., FinFET and GAAFET). This selection aligns with academic benchmarks, as planar device designs frequently reference dimensions within the 22–30 nm range to maintain simulation reliability and interpretability. Meanwhile, the 14 nm L_G was chosen to assess the scalability of the GAAFET PTnMOSFET structure. This choice reflects dimensions typical of advanced node technologies, enabling direct comparisons with state-of-the-art designs (e.g., Fig. 7 in [29]) and demonstrating the robustness of PTnMOSFET performance under aggressive scaling.

In all cases, the L_G , L_{Spacer} , L_{Source} , and L_{Drain} along the x-direction are greater than 10 nm, leading to a uniform mesh size of 1 nm in the x-direction. In the y-direction, a finer mesh size of 0.1 nm is used for the gate oxide layer to ensure simulation accuracy, while the buried oxide layer, with a thickness of 50 nm, has minimal impact on punch-through current and uses a 10 nm mesh size. For FinFET and GAAFET, the z-direction mesh size of the gate oxide layer is set to 0.1 nm, and the fin width (W_{Fin}) and nanosheet width (W_{NS}) are both 10 nm, with a mesh size of 1 nm.

To ensure realistic and accurate simulations, we performed calibrations for the Planar, FinFET, and GAAFET structures using Silvaco Athena [16] and Atlas [17] (as shown in Fig. 4). This process involved adjusting mobility parameters for PTnMOSFET and nMOSFET based on the planar, FinFET, and GAAFET nMOSFET structures referenced in [18, 19], and [20], with all models calibrated against experimental results to accurately represent device performance. For accuracy, the device dimensions and doping concentrations used for calibration were identical to those in the reference materials.

Fig. 3 (a) GAAFET SC CMOS Proposed and its cross-sectional views along **(b)** A-A' plane and **(c)** B-B' plane, respectively

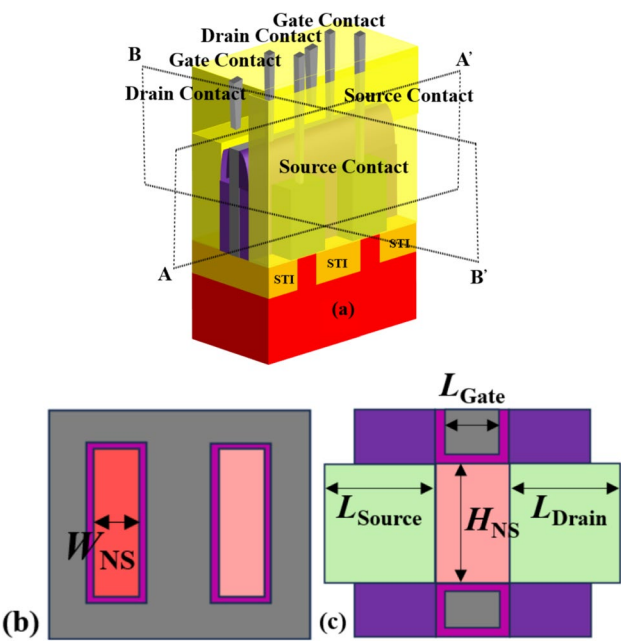


Table 1 Device parameters used for planar PTnMOSFET

Device parameters	
Channel Thickness (t_{si})	10 nm
Gate Length (L_G)	30 nm and 14 nm
Source Length (L_{Source})	20 nm
Drain Length (L_{Drain})	20 nm
Spacer Length (L_{Spacer})	20 nm
Gate Oxide Thickness (t_{ox})	3 nm
Channel Doping (p ⁻ -type)	$1 \times 10^{17} \text{ cm}^{-3}$
Source/Drain Doping (n ⁻ -type)	$2 \times 10^{17} \text{ cm}^{-3}$
Channel Material	Silicon
Area [$W (L_{Source} + 2L_{Spacer} + L_G + L_{Drain})$]	1100nm ²

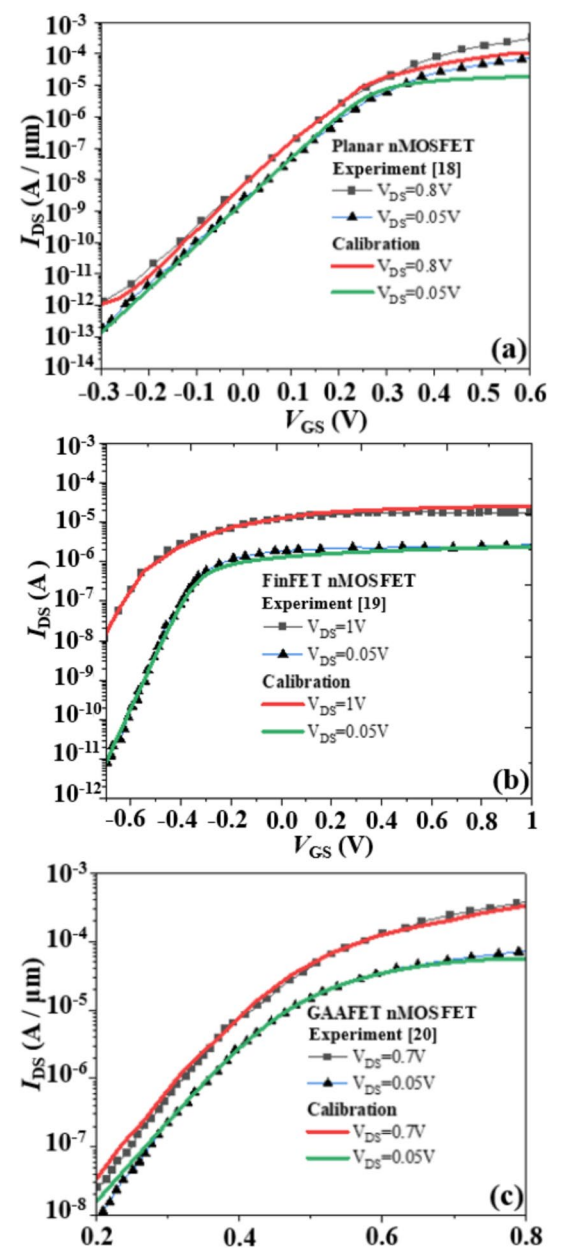
Table 2 Device parameters used for FinFET PTnMOSFET

Device parameters	
Fin Height (H_{Fin})	90 nm
Fin Width (W_{Fin})	10 nm
Gate Length (L_G)	30 nm and 14 nm
Source Length (L_{Source})	40 nm
Drain Length (L_{Drain})	40 nm
Gate Oxide Thickness (t_{ox})	3 nm
Channel Doping (p ⁻ -type)	$1 \times 10^{17} \text{ cm}^{-3}$
Source/Drain Doping (n ⁻ -type)	$2 \times 10^{17} \text{ cm}^{-3}$
Channel Material	Silicon
Area [$W_{Fin} (L_{Source} + L_G + L_{Drain})$]	1100nm ²

The calibration also included multi-bias settings, specifically calibrating the device models at $V_{DS} = 0.05 \text{ V}$ and higher V_{DS} values, to capture current behavior across a range of V_{DS} . This multi-bias approach ensures that our PTnMOSFET model reliably represents performance across a comprehensive operating range. Additionally, various physical effects were considered: carrier generation and recombination (with SRH [21], CONSRH [22–24], and AUGER

Table 3 Device parameters used for GAAFET PTnMOSFET

Device parameters	
Nanosheet Height (H_{NS})	90 nm
Nanosheet Width (W_{NS})	10 nm
Gate Length (L_G)	30 nm and 14 nm
Source Length (L_{Source})	40 nm
Drain Length (L_{Drain})	40 nm
Gate Oxide Thickness (t_{ox})	3 nm
Channel Doping (p ⁻ -type)	$1 \times 10^{17} \text{ cm}^{-3}$
Source/Drain Doping (n ⁻ -type)	$2 \times 10^{17} \text{ cm}^{-3}$
Channel Material	Silicon
Area [$W_{NS}(L_{Source} + L_G + L_{Drain})$]	1100nm ²

Fig. 4 Calibration plots of simulation parameters for PTnMOSFET devices: **(a)** Planar PTnMOSFET, **(b)** FinFET PTnMOSFET, and **(c)** GAAFET PTnMOSFET

[25, 26] models), impact ionization (IMPACT SELB), gate leakage (FNORD, FNHOLES, and BBT.KL), and carrier mobility (CONMOB, FLDMOB, and CVT to account for effects from doping concentration, vertical electric field, parallel electric field, and temperature). Higher doping effects were modeled with Fermi–Dirac statistics and bandgap narrowing (BGN) models.

3 Device fabrication

In Fig. 5, the proposed GAAFET PTnMOSFET follows the general GAA fabrication process [27], modeled and optimized in Silvaco Athena with a simulation temperature set at 300 K. The doping concentration of the source and drain was precisely adjusted to fall within a design window that facilitates controlled punch-through. The fabrication process begins with multilayer SiGe/Si stacks grown on the wafer, Vertical fins were patterned through advanced reactive ion etching (RIE) to achieve the desired fin height and profile. Next, SiO₂ was deposited to a thickness of 145 nm, followed by chemical mechanical planarization (CMP) for surface leveling. SiO₂ etching was then conducted to expose the fin and form shallow trench isolation (STI). A dummy gate stack was constructed on the fin, and doping was introduced with specified concentrations of 2×10^{17} (cm⁻³) for light doping in the PTnMOSFET's source/drain and 1×10^{18} (cm⁻³) for heavy doping in the nMOSFET's source/drain. Source/drain epitaxy, designed to minimize parasitic resistance, was performed. A photoresist (PR) coating, exposure, and development were applied to define the spacer region accurately. Si₃N₄ spacers were formed using deposition and RIE. Interlayer dielectrics (ILDs) were deposited with a CMP to ensure planarity, and the dummy gate was removed by chemical wet etching in a dilute hydrofluoric (DHF) solution. Si nanosheet channels were released via selective wet etching of SiGe layers. Following this, HfO₂ with a thickness of 3 nm and metal gate electrodes were deposited through atomic layer deposition (ALD) to complete the GAA structure. The entire GAAFET SC CMOS device was completed with back-end metallization interconnection to finalize the layout.

Fig. 5 Fabrication process flow for the proposed GAAFET single-carrier (SC) CMOS

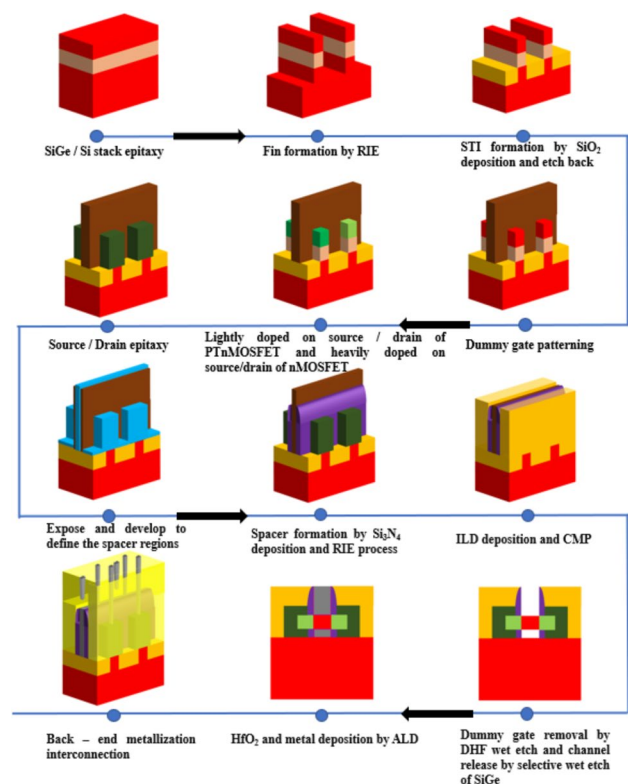


Fig. 6 Operating states of the GAAFET PTnMOSFET: **(a)** $V_G \leq V_{FB}$, **(b)** $V_{FB} \leq V_G \leq V_{TH}$

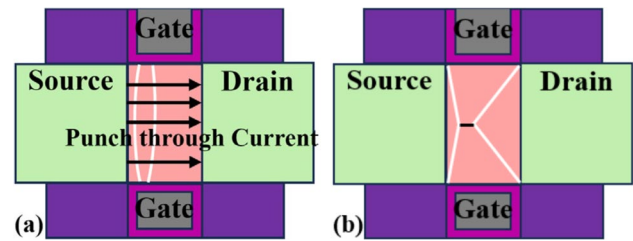
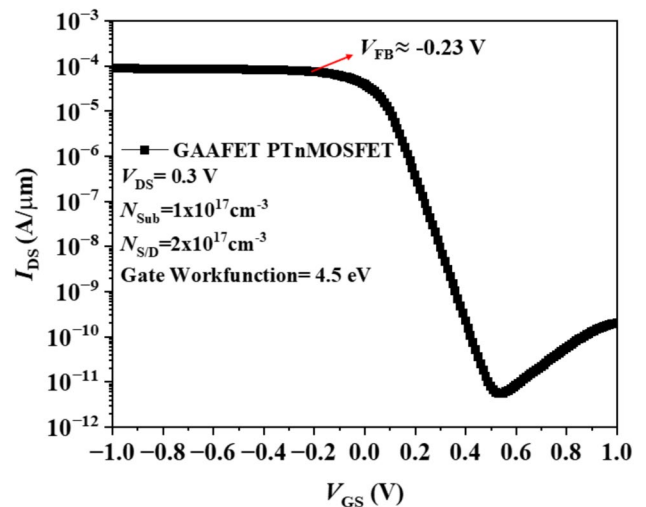


Fig. 7 Subthreshold Characteristics of GAAFET PTnMOSFET



4 Device operation mechanism

Figures 6 and 7 illustrate the unique operation mechanism of our PTnMOSFET, which operates through punch-through current rather than the conventional inversion layer current seen in traditional nMOSFETs and pMOSFETs. In our PTnMOSFET, a gate work function of 4.5 eV is set to achieve an appropriate flat-band voltage (V_{FB}). In Fig. 6, the white lines represent the depletion regions of the source and drain, while the black line denotes the gate depletion region. Based on Fu-Chieh Hsu's theories [28], the device operates in different modes depending on the gate voltage (V_G) relative to V_{FB} . In Fig. 6a, when V_G is below V_{FB} , the device operates in accumulation mode. At this point, the punch-through effect occurs beneath the surface, allowing electrons to flow between the source and drain at the potential minimum, resulting in a fully on-state. This mode resembles the saturation region where the device conducts fully without forming an inversion layer. In Fig. 6b, as V_G increases beyond V_{FB} but remains below the threshold voltage (V_{TH}), the device enters a weak inversion mode. The gate depletion region expands and pushes the source and drain depletion regions apart, reducing the punch-through effect and transitioning the device toward the off-state. This intermediate mode does not correspond directly to the traditional triode or saturation regions but is characterized by a reduction in current flow. As shown in Fig. 7, V_{FB} (approximately -0.23 V) acts similarly to V_{TH} in a conventional pMOSFET. When V_G is below V_{FB} , the gate does not create a depletion region, keeping the device fully conductive. Conversely, when V_G exceeds V_{FB} , a depletion region forms, isolating the source and drain and effectively turning off the device.

In summary, our PTnMOSFET operates based on punch-through current rather than following the typical linear (ohmic) or saturation region behaviors. The source and drain depletion regions connect when their doping concentrations are near the channel's concentration, allowing electron punch-through without forming an inversion layer.

5 Results and discussion

In a previous study [29], we proposed using silicon as the material for the PTnMOSFET. However, prior research on the design window for punch-through phenomena in PTnMOSFETs has not been sufficiently comprehensive, and optimal transfer curves and subthreshold characteristics were not achieved at that time. In response, this study introduces planar, FinFET, and GAAFET technologies in conjunction with PTnMOSFET devices to facilitate a more in-depth investigation.

5.1 PTnMOSFET design window comparison of three structures

Figure 8 illustrates the logarithm of the drain current ($\log(I_{DS})$) as a function of the gate-source voltage (V_{GS}), with a drain-source voltage (V_{DS}) set at 0.3 V for planar (Fig. 8a), FinFET (Fig. 8b), and GAAFET (Fig. 8c) PTnMOSFETs. The results indicate that when the source/drain doping concentration is sufficiently close to the channel concentration (2 to 4 times greater

Fig. 8 Subthreshold characteristics of PTnMOSFETs with varying source/drain doping concentrations: **(a)** Planar PTnMOSFET, **(b)** FinFET PTnMOSFET, and **(c)** GAAFET PTnMOSFET

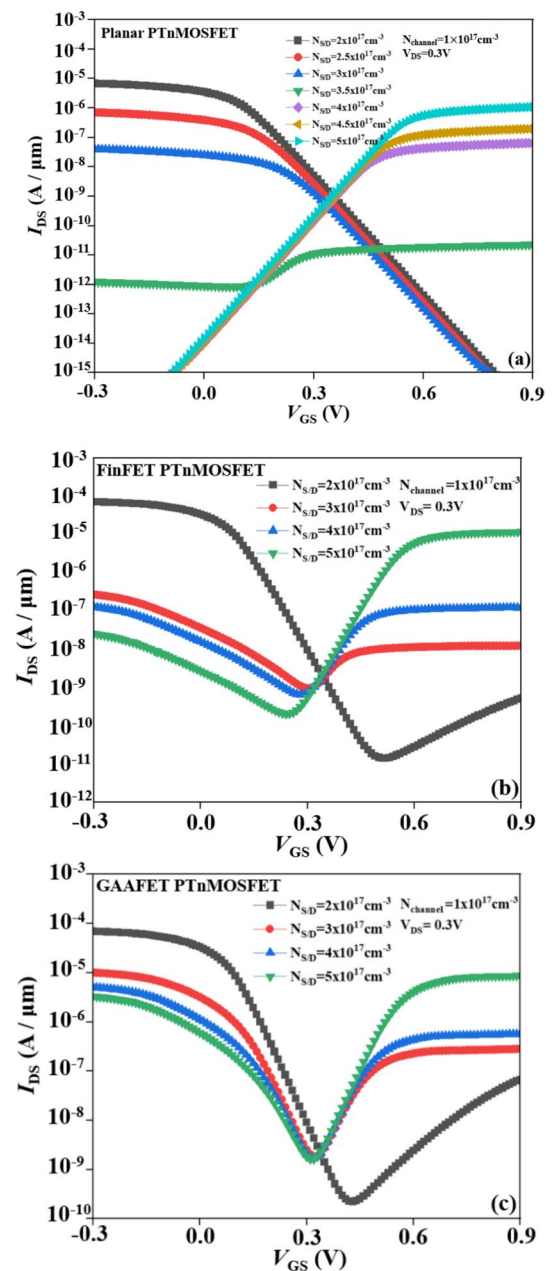


Fig. 9 Design windows for Planar, FinFET, and GAAFET PTnMOSFET structures

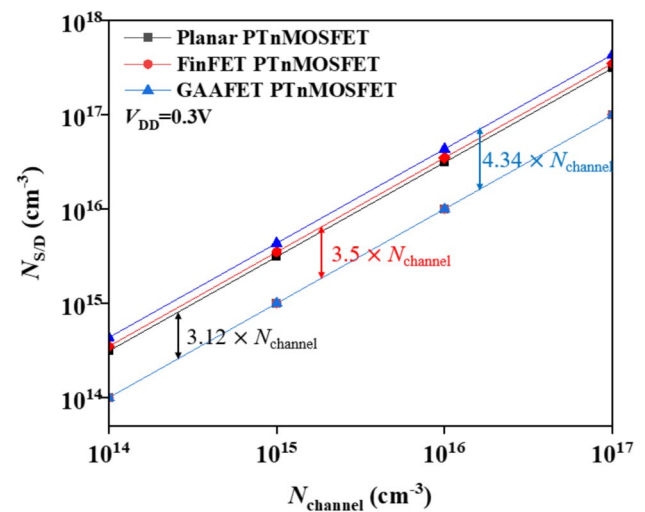
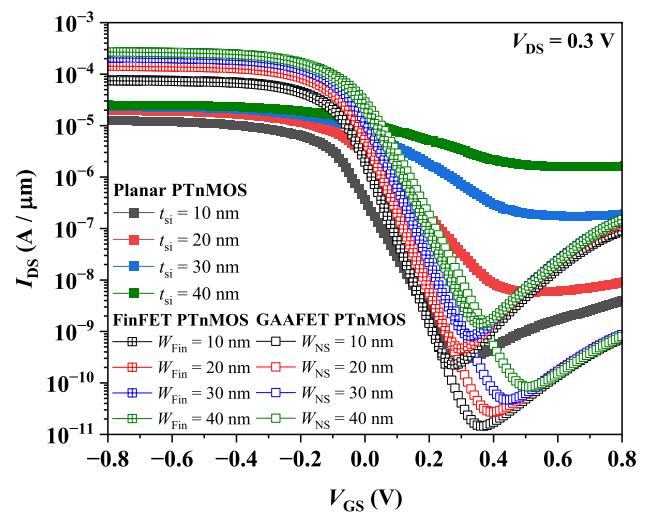


Fig. 10 Subthreshold characteristics of PTnMOSFETs with modulation of t_{si} , W_{Fin} , and W_{NS} for Planar, FinFET, and GAAFET structures



than the channel concentration), the PTnMOSFET demonstrates optimal performance, resulting in the highest I_{ON} and $I_{DS}-V_{GS}$ characteristics similar to those of a pMOSFET. This is because, under these conditions, electrons can easily punch through from the source to the drain, generating significant punch-through current. However, when the source/drain doping concentration exceeds the design window, the $I_{DS}-V_{GS}$ characteristics of the PTnMOSFET start to resemble those of an nMOSFET. We refer to the range of source/drain doping concentrations that enable $I_{DS}-V_{GS}$ characteristics similar to those of a pMOSFET as the "PTnMOSFET design window."

The two blue curves in Fig. 9 represent the upper and lower bounds of the design window for the GAAFET PTnMOSFET. Each structure (Planar, FinFET, and GAAFET PTnMOSFET) has two lines that outline different doping concentration limits. The area between these lines indicates the range where the source/drain doping concentration should be set to ensure that the depletion regions in the channel make contact, which is necessary for forming a PTnMOSFET.

For the GAAFET structure specifically, the two blue curves mark this design window's limits, highlighting the viable range for doping concentrations. This visual representation helps illustrate the conditions under which the PTnMOSFET can effectively operate.

5.2 Optimized electrical characteristics of three PTnMOSFET structures

Figure 10 shows the subthreshold characteristics of PTnMOSFETs with modulation of t_{si} , W_{Fin} , and W_{NS} for Planar, FinFET, and GAAFET structures. In planar PTnMOSFET, when t_{si} is 40 nm, I_{ON} reaches 1.4×10^{-5} (A/μm), whereas I_{OFF} reaches 3.38×10^{-6} (A/μm). As t_{si} decreases to 10 nm, I_{ON} decreases from 1.4×10^{-5} (A/μm) to 1.29×10^{-5} (A/μm). Fortunately, I_{ON}/I_{OFF}

overall improved because the I_{OFF} decreased from 3.38×10^{-6} (A/ μm) to 3.15×10^{-10} (A/ μm). Increasing t_{si} enlarges the punch through area, thereby increasing the punch through current. However, when the channel thickness increases, the gate must apply a higher voltage to ensure that the depletion region controlled by the gate fully occupies the channel. This makes the device harder to turn off, resulting in a large I_{OFF} . In FinFET and GAAFET PTnMOSFET, the high aspect ratio of $W_{\text{Fin}}:H_{\text{Fin}} = W_{\text{NS}}:H_{\text{NS}} = 1:9$ causes the gate electric fields parallel to W_{Fin} and the W_{NS} direction to dominate the device's control. Consequently, increasing W_{Fin} and W_{NS} enlarges the punch through the area and boosts the I_{ON} but also makes the device more challenging to turn off, which is similar to the planar PTnMOSFET. This leads to an increased leakage current and a poor $I_{\text{ON}}/I_{\text{OFF}}$. Conversely, reducing W_{Fin} and W_{NS} is good for $I_{\text{ON}}/I_{\text{OFF}}$ improvement, decreasing I_{OFF} and saving wafer area, but it results in a lower I_{ON} .

Figure 11 shows the subthreshold characteristics for the optimization of H_{Fin} and H_{NS} in FinFET and GAAFET PTnMOSFETs. The results show that as H_{Fin} and H_{NS} increase from 60 to 90 nm, the I_{ON} of the FinFET PTnMOSFET increases from 5.1×10^{-5} (A/ μm) to 8.6×10^{-5} (A/ μm), and the I_{ON} of the GAAFET PTnMOSFET increases from 5.2×10^{-5} (A/ μm) to 8.7×10^{-5} (A/ μm). Fortunately, because the increase in I_{OFF} is not as significant as the increase in I_{ON} is, the overall $I_{\text{ON}}/I_{\text{OFF}}$ still improves. We learn that larger values of H_{Fin} and H_{NS} increase the punch-through area, thereby increasing the punch-through current, which leads to a higher I_{ON} . Furthermore, the gate control capabilities of FinFET and GAAFET PTnMOSFET are also enhanced, resulting in an improvement in the SS_{avg} . Notably, when planar, FinFET, and GAAFET PTnMOSFET occupy the same wafer area, increasing H_{Fin} and H_{NS} enables FinFET and GAAFET PTnMOSFET to achieve better SS_{avg} , I_{ON} and $I_{\text{ON}}/I_{\text{OFF}}$ values than planar PTnMOSFET.

In Fig. 12a, the purpose of $I_{\text{ON(PTnMOSFET)}}/I_{\text{ON(nMOSFET)}}$ is to ensure that PTnMOSFET achieves an I_{ON} similar to that of nMOSFET while occupying the same wafer area. This finding indicates that PTnMOSFET can achieve current matching without the need for increasing width or introducing stress techniques, unlike traditional pMOSFET. Instead, we utilized the punch-through mechanism to bring the $I_{\text{ON(PTnMOSFET)}}$ close to the $I_{\text{ON(nMOSFET)}}$. GAAFET PTnMOSFET achieves an $I_{\text{ON(PTnMOSFET)}}/I_{\text{ON(nMOSFET)}} = 0.89$, which is better than $I_{\text{ON(PTnMOSFET)}}/I_{\text{ON(nMOSFET)}} = 0.84$ (Planar PTnMOSFET) in the same wafer area. This is attributed to the large H_{NS} , which enlarges the punch-through area, leading to a larger punch-through current (higher I_{ON}). Device parameters for planar, FinFET, and GAAFET PTnMOSFET and nMOSFET are listed in Fig. 12b. In Fig. 13, we calculate SS_{avg} by averaging across three to four decades. The SS_{avg} of the GAAFET PTnMOSFET is 74 mV/dec, which is the best among the three PTnMOSFET devices. This superior performance makes GAAFET PTnMOSFET ideal for applications in CMOS and SRAM circuits, as mentioned in parts D and F.

5.3 DIBL—like effect on 3 PTnMOSFET structures

In Fig. 14a, the V_{DD} variation analysis of planar, FinFET, and GAAFET nMOSFET devices are shown under $V_{\text{DD}} = 0.1$ V, 0.2 V, and 0.3 V. As V_{DD} increases, the depletion regions controlled by the drain and source terminals gradually approach each other, leading to the DIBL effect, which may cause conventional devices to operate abnormally. For the planar, FinFET, and GAAFET nMOSFETs, we calculated the DIBL values to be 100, 50, and 50 mV/V, respectively,

Fig. 11 Subthreshold characteristics for the optimization of H_{Fin} and H_{NS} in FinFET and GAAFET PTnMOSFETs

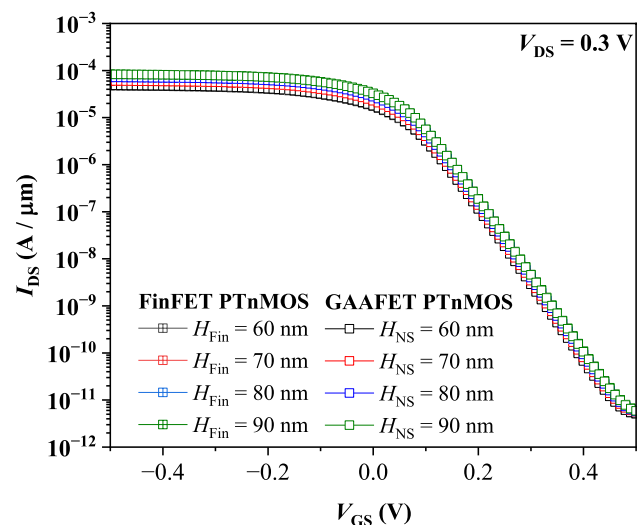


Fig. 12 (a)

$I_{ON(PTnMOSFET)}/I_{ON(nMOSFET)}$ comparison among planar, FinFET, and GAAFET structures. **(b)** Device parameters for planar, FinFET, and GAAFET PTnMOSFET and nMOSFET

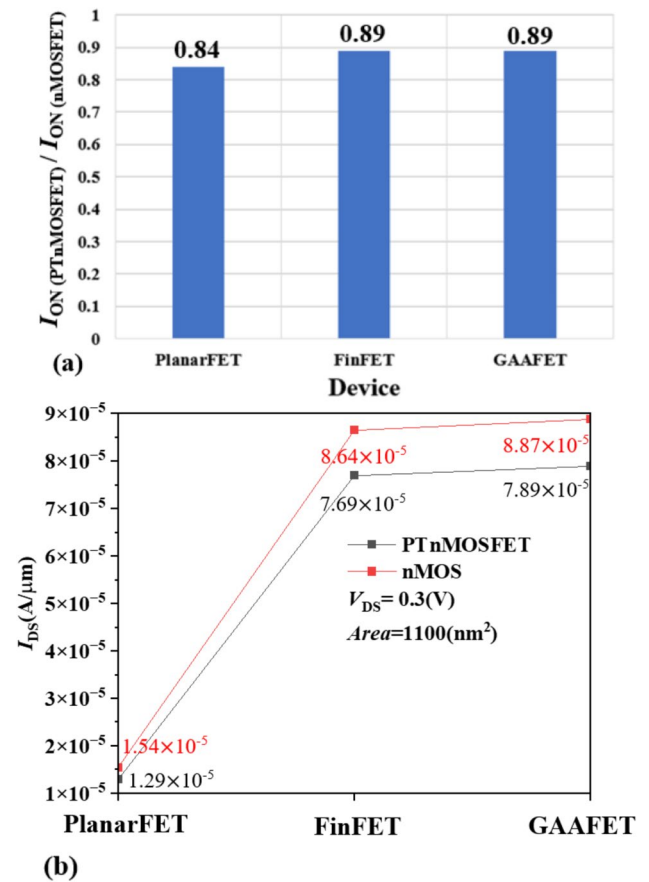
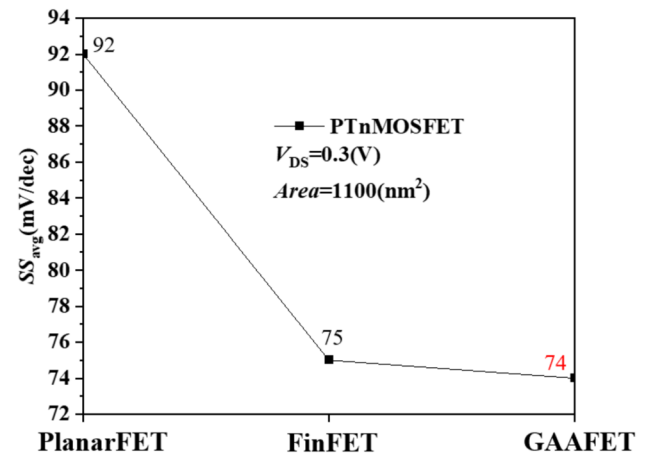


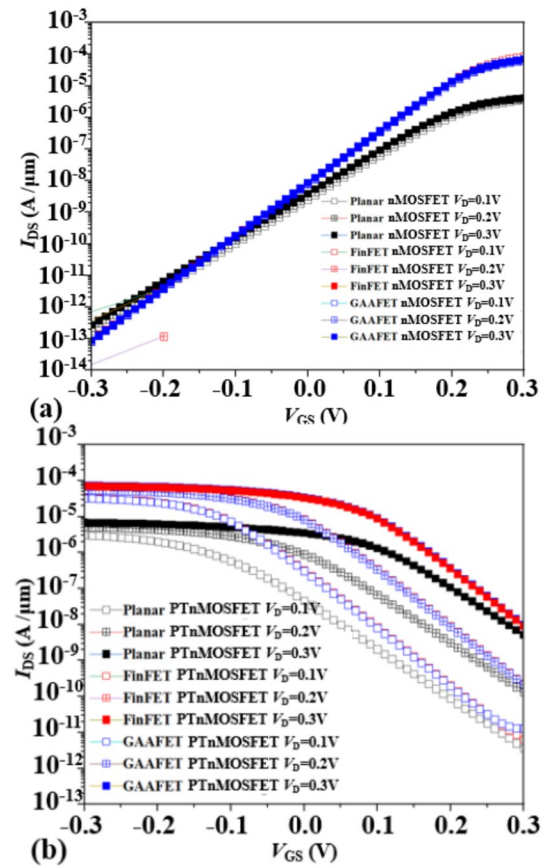
Fig. 13 Comparison of SS_{avg} for Planar, FinFET, and GAAFET PTnMOSFET structures



with the GAAFET nMOSFET showing the lowest DIBL due to its superior gate control, which effectively minimizes the impact of V_{DD} variations on the channel.

In contrast, Fig. 14b presents the V_{DD} variation analysis of planar, FinFET, and GAAFET PTnMOSFET devices under $V_{DD} = 0.1 V, 0.2 V$, and $0.3 V$). By intentionally adjusting the doping concentrations in the source, channel, and drain regions, we induce the DIBL-Like effect to produce a punch-through current, which serves as the conduction current for PTnMOSFET devices. Unlike conventional MOSFETs, where DIBL effects are often problematic due to short channel lengths, in PTnMOSFET devices, the DIBL-Like effect is attributed to the high sensitivity of the punch-through current to V_{DD} . This high sensitivity enhances the charging and discharging response in CMOS circuits, improving performance in these applications.

Fig. 14 V_{DD} variation analysis for planar, FinFET, and GAAFET structures: **(a)** nMOSFET and **(b)** PTnMOSFET



Our PTnMOSFET devices operate based on the non-destructive punchthrough behavior between the depletion regions controlled by the source and drain, respectively. Due to this unique mechanism, the encroachment controlled by the drain is significant, which results in a larger DIBL-like value. This characteristic is particularly advantageous for PTnMOSFETs as they are further scaled down, supporting better performance at reduced sizes.

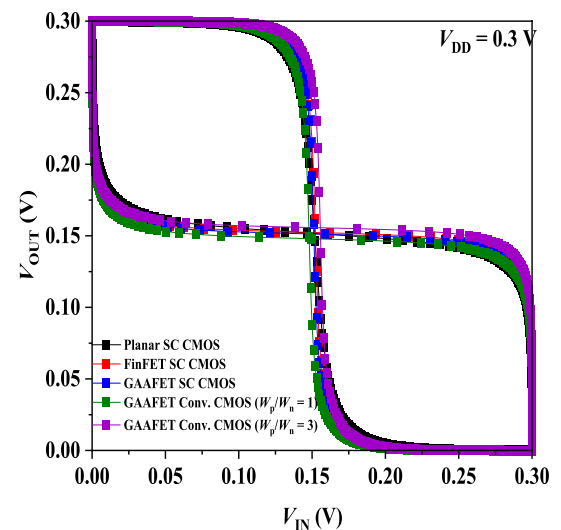
As semiconductor technology scales beyond 3 nm, addressing the impact of quantum mechanical tunneling and pronounced SCEs becomes essential. Traditional nMOSFETs and pMOSFETs experience significant V_{TH} variations due to these scaling-induced phenomena, affecting device stability and performance. However, our PTnMOSFET design circumvents this issue through the use of V_{FB} as the defining parameter for threshold behavior. Unlike conventional V_{TH} , V_{FB} remains unaffected by changes in device length and width, ensuring inherent stability against V_{TH} roll-off. This unique property allows PTnMOSFETs to maintain consistent performance even as device dimensions shrink.

Furthermore, when the V_G is below V_{FB} , electron punch-through occurs, generating a punch-through current that places the PTnMOSFET in the on-state. As V_G surpasses V_{FB} , the gate depletion region gradually extends to occupy the channel, blocking the punch-through current and turning the device off. This behavior highlights V_{FB} as a more suitable threshold parameter, supporting stable device operation amid the challenges posed by continued scaling and quantum mechanical effects.

5.4 Comparison of Conv. and SC devices on CMOS circuit

In above Part B, the $I_{ON(PTnMOSFET)}/I_{ON(nMOSFET)}$ for the three structures is significantly greater than 0.33 but not equal to 1. Therefore, in the circuit analysis, a CMOS circuit with $W_p/W_n = 3$ was included as a control group to observe the performance differences between the SC CMOS and Conv. CMOS ($W_p/W_n = 3$), where the latter exhibits superior performance to the SC CMOS. We used Silvaco MixMode simulations to analyze the SC CMOS (composed of PTnMOSFET and nMOSFET) between planar, FinFET, and GAAFET structures. The device with the best performance among the three devices was subsequently selected and compared with Conv. CMOS ($W_p/W_n = 1$) and Conv. CMOS ($W_p/W_n = 3$). Figure 15 depicts the butterfly curves for conventional CMOS ($W_p/W_n = 1$), ($W_p/W_n = 3$), and SC CMOS. The GAAFET structure exhibits the best

Fig. 15 Butterfly curves for conventional CMOS ($W_p/W_n=1$), ($W_p/W_n=3$), and SC CMOS



I_{ON} and SS_{avg} , leading to higher NM_H and NM_L values than those of the planar device. Moreover, owing to the mismatch in I_{ON} between nMOSFET and pMOSFET ($I_{ON(pMOSFET)}/I_{ON(nMOSFET)} \approx 0.33$), the NM_H and NM_L values of Conv. The CMOS ($W_p/W_n=1$) differs significantly for the GAAFET structure. In a SC CMOS, replacing pMOSFET with PTnMOSFET allows I_{ON} to approach matching ($I_{ON(PTnMOSFET)}/I_{ON(nMOSFET)} \approx 0.89$) without increasing process costs, resulting in NM_H and NM_L values that are more comparable than Conv. CMOS ($W_p/W_n=1$).

Figure 16 shows the NM_H and NM_L as a percentage of V_{DD} for conventional CMOS ($W_p/W_n=1$), ($W_p/W_n=3$), and SC CMOS, which indicates that the NM_H and NM_L values of the GAAFET SC CMOS are more comparable than those of Conv. CMOS ($W_p/W_n=1$). Figure 17 shows the voltage gain comparison for conventional CMOS ($W_p/W_n=1$), ($W_p/W_n=3$), and SC CMOS. Due to the higher I_{ON} of PTnMOSFET, the switching speed of PTnMOSFET is faster than that of pMOSFET ($W_p/W_n=1$). Therefore, the SC CMOS has a larger maximum gain than the Conv. CMOS ($W_p/W_n=1$). Figure 18 presents the I_{OUT} - V_{IN} characteristic curves for conventional CMOS ($W_p/W_n=1$), ($W_p/W_n=3$), and SC CMOS, where the area under the curves represents the static power consumption of the CMOS devices. The static power dissipation of the GAAFET Conv. The CMOS ($W_p/W_n=3$) is the highest among all devices. Conversely, the planar SC CMOS ($W_p/W_n=1$) has the lowest power consumption. This is because the I_{ON} value significantly affects the power consumption of the device. Figure 19 shows the transient response times for conventional CMOS ($W_p/W_n=1$), ($W_p/W_n=3$), and SC CMOS. The results indicate that the SC devices, while having a minimal footprint, show competitive delay times compared to other inverters. Although their delay is slightly larger than that of a fully-sized conventional GAAFET, it is important to note that the conventional GAAFET requires significantly more wafer area. This comparison highlights the trade-off between device footprint and performance in delay time. All the characteristic values mentioned in part D are shown in Table 4.

Fig. 16 NM_H and NM_L as a percentage of V_{DD} for conventional CMOS ($W_p/W_n=1$), ($W_p/W_n=3$), and SC CMOS

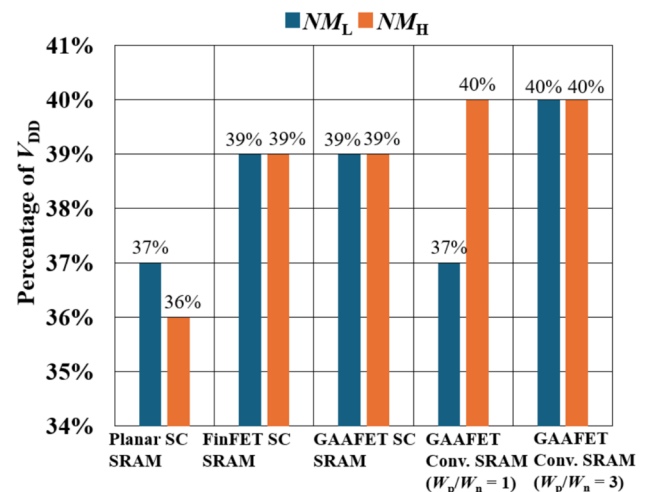


Fig. 17 Voltage gain comparison for conventional CMOS ($W_p/W_n=1$), ($W_p/W_n=3$), and SC CMOS

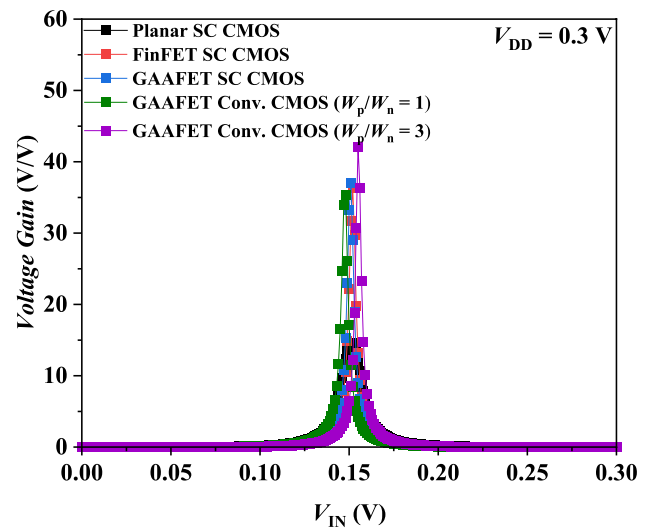


Fig. 18 I_{OUT} - V_{IN} curves for conventional CMOS ($W_p/W_n=1$), ($W_p/W_n=3$), and SC CMOS

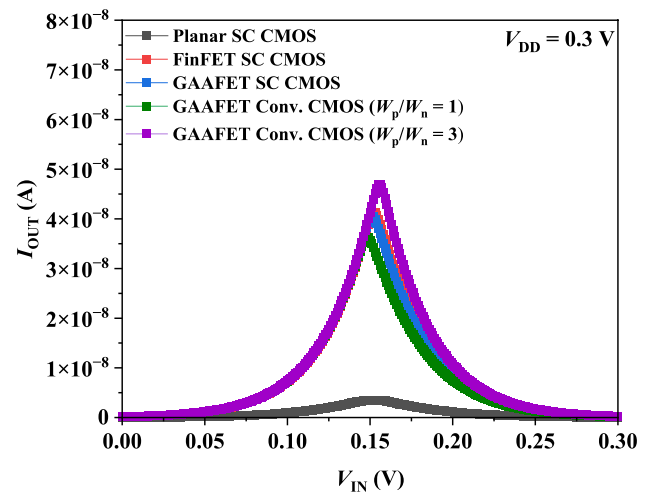


Fig. 19 Transient response times for conventional CMOS ($W_p/W_n=1$), ($W_p/W_n=3$), and SC CMOS

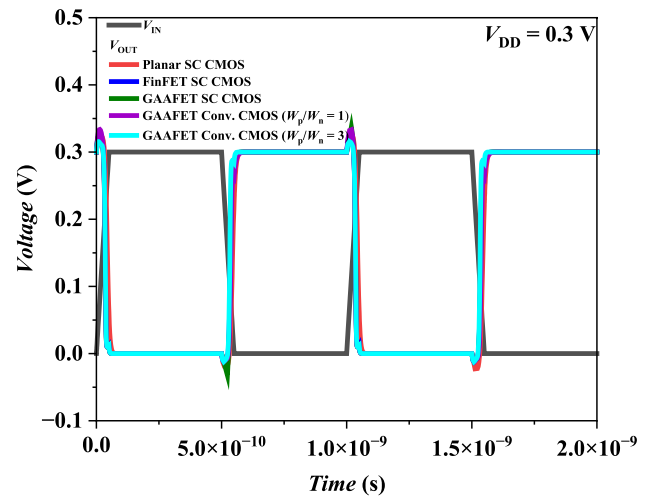


Table 4 Performance Comparison of SC CMOS and 6 T SC SRAM with Conventional Counterparts

$V_{DD} = 0.3$ (V)		Single-Carrier			Conv. ($W_p/W_n = 1$)	Conv. ($W_p/W_n = 3$)
		Planar	FinFET	GAAFET	GAAFET	GAAFET
CMOS	NM_H (mV)	108	118	117	120	120
	NM_L (mV)	110	116	118	113	121
	Gain	15	36	37	35	42
	Power Dissipation (W)	$2.91e^{-10}$	$2.55e^{-9}$	$2.62e^{-9}$	$2.33e^{-9}$	$2.88e^{-9}$
	Propagation Delay (s)	$1.9e^{-11}$	$1.05e^{-11}$	$1e^{-11}$	$1.18e^{-11}$	$9.1e^{-12}$
SRAM	SNM_{Hold} (mV)	109	117	118	116	121
	SNM_{Read} (mV)	50	55	56	52	59
	SNM_{Write} (mV)	140	145	147	144	153

5.5 Robustness in design and fabrication

Our proposed GAAFET PTnMOSFET and GAAFET SC CMOS designs offer notable robustness in both design and fabrication processes. The PTnMOSFET operates at a lower drain bias and exhibits $I_{DS}-V_{GS}$ characteristics similar to those of a pMOSFET due to its non-destructive punch-through effect. This makes it highly compatible with existing nMOSFET fabrication processes. Importantly, PTnMOSFET can be realized by simply adjusting the source/drain doping concentrations, requiring no additional process steps and facilitating seamless integration within traditional nMOSFET process flows.

The GAAFET SC CMOS design provides further robustness by eliminating the need for complex transistor sizing and relying exclusively on electron transport. This approach addresses carrier mismatch issues, significantly reduces fabrication costs, and minimizes wafer area usage. Additionally, this design demonstrates a high noise margin, a symmetric butterfly curve, and superior voltage gain and delay time when compared to conventional CMOS. These attributes ensure that the GAAFET SC CMOS is both robust and reliable, supporting performance stability and cost-efficiency, which are critical for advanced applications.

5.6 Comparison of Conv. and SC devices in SRAM circuits

Figure 20 presents the circuit diagram of our proposed GAAFET SC 6 T SRAM. Figure 21a–c display the butterfly curves for the hold mode, read mode and write mode respectively, comparing the 6 T conventional SRAM with $W_p/W_n = 3$, SRAM with $W_p/W_n = 1$, and the SC SRAM across planar, FinFET, and GAAFET structures. The SNM_{Hold} , SNM_{Read} and SNM_{Write} values are shown in Table 4. The simulation results demonstrate that the GAAFET SC SRAM has larger SNM values in the hold mode, read mode and write mode than any SC SRAM. This is attributed to the better SS_{avg} and I_{ON} of the GAAFET structure. In Table 4, the performance comparison of SC CMOS and 6 T SC SRAM with conventional Counterparts are presented. The table shows that replacing pMOSFET ($W_p/W_n = 1$) with PTnMOSFET in both CMOS and SRAM circuits leads to improved characteristics while maintaining the same wafer area, thus increasing the circuit speed and stability while reducing the fabrication costs.

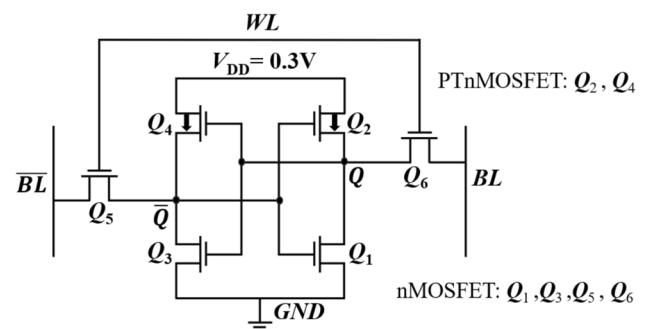
Fig. 20 Circuit diagram of the GAAFET SC 6 T SRAM

Fig. 21 Butterfly curves for various 6 T SRAMs in (a) hold mode (b) read mode and (c) write mode

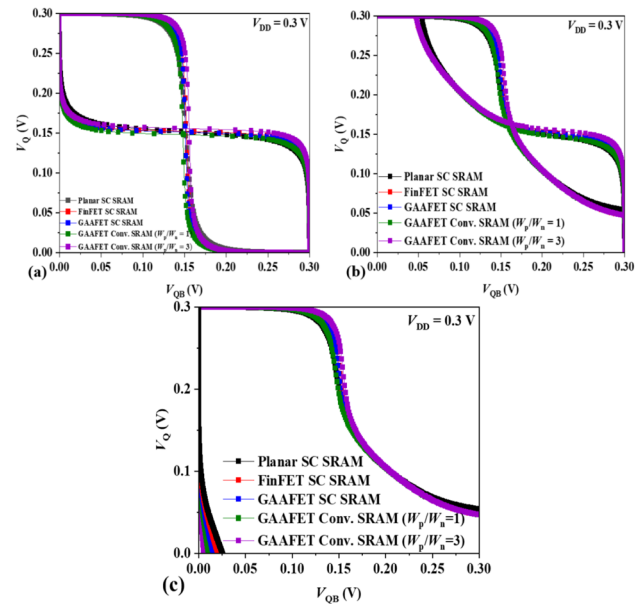
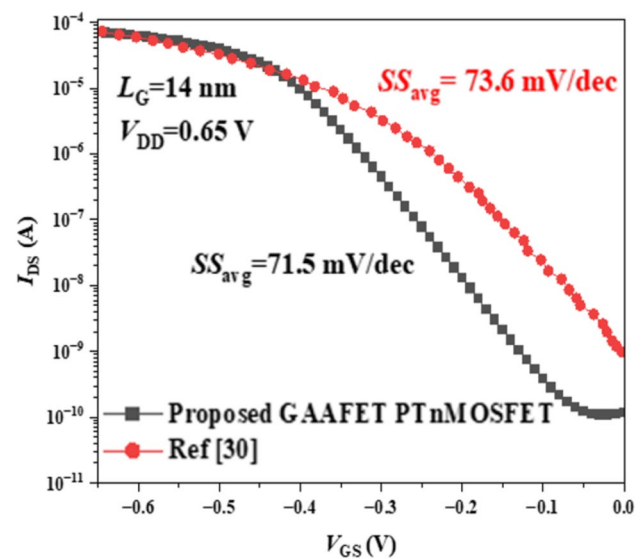


Fig. 22 Subthreshold characteristics ($\text{Log}(I_{DS})$) as a function of V_{GS} for sub-5 nm technology node with $L_G = 14$ nm and $V_{DS} = 0.65$ V



5.7 Discuss results at 14 nm gate length

Figure 22 illustrates the simulation results of the GAAFET PTnMOSFET compared with the device characteristics reported in Fig. 7 of [30], under identical conditions with the L_G of 14 nm and a V_{DD} of 0.65 V. The selection of 14 nm as the L_G reflects its relevance to advanced node technologies and facilitates direct benchmarking against state-of-the-art devices. The simulation results reveal that the GAAFET PTnMOSFET achieves a SS_{avg} of 71.5 mV/dec, demonstrating its ability to maintain competitive performance under aggressive scaling.

This analysis highlights the scalability of the PTnMOSFET architecture and its capability to deliver superior subthreshold characteristics and current matching. The robust performance of the PTnMOSFET, underscores its potential for ultra-scaled CMOS and SRAM applications. These findings validate the PTnMOSFET as a promising solution for meeting the stringent performance and scaling requirements of next-generation semiconductor technologies.

5.8 Benchmark

The minimum operational V_{DD} for SRAM, often termed the read failure voltage, is the point at which the read signal noise margin (SNM_{Read}) becomes insufficient to maintain stable read operations. This value is typically defined as the voltage at which the SNM_{Read} equals the thermal noise margin, indicating the limit below which read stability deteriorates.

In our study, we performed comprehensive simulations to determine the read failure voltage for different SRAM structures, including our proposed GAAFET SC SRAM, conventional CFET SRAM [30], and traditional 6 T SRAM [31]. The process involved simulating the SNM_{Read} across various V_{DD} levels (0.2 V, 0.25 V, 0.3 V, 0.35 V, and 0.4 V) for each structure. The results were used to plot a trend showing how SNM_{Read} varies with V_{DD} , as depicted in Fig. 23a of the revised manuscript. By extrapolating the plotted trends for each SRAM structure, we identified the respective minimum V_{DD} values at which read stability begins to deteriorate. These values were found as 145 mV for the proposed GAAFET SC SRAM, 176 mV for the conventional CFET SRAM, and 190 mV for the 6 T SRAM. These minimum V_{DD} values indicate the enhanced low-voltage performance of our GAAFET SC SRAM compared to conventional CFET and 6 T SRAM structures. Figure 23b summarizes these read failure voltage points, demonstrating that the proposed GAAFET SC SRAM can maintain read stability at a significantly lower V_{DD} , underscoring its potential for ultra-low-power applications.

Incorporating Process, Voltage, and Temperature (PVT) variation analyses is crucial to understanding the full performance range of PTnMOSFETs. These analyses help predict device behavior under real-world conditions, where variations in temperature or voltage can significantly impact circuit reliability [32]. Developing robust models that account for these variations ensures that PTnMOSFET-based circuits meet the reliability standards necessary for next-generation applications. While our initial simulations were conducted at a constant temperature of 300 K within a controlled process environment, we acknowledge that this analysis did not incorporate full PVT variations. We have noted this limitation and suggest that future work should include comprehensive PVT variation analyses to better assess the robustness and reliability of PTnMOSFETs in practical applications.

Fig. 23 (a) Read signal noise margin at different V_{DD} values and (b) read failure voltage comparison for various SRAM structures

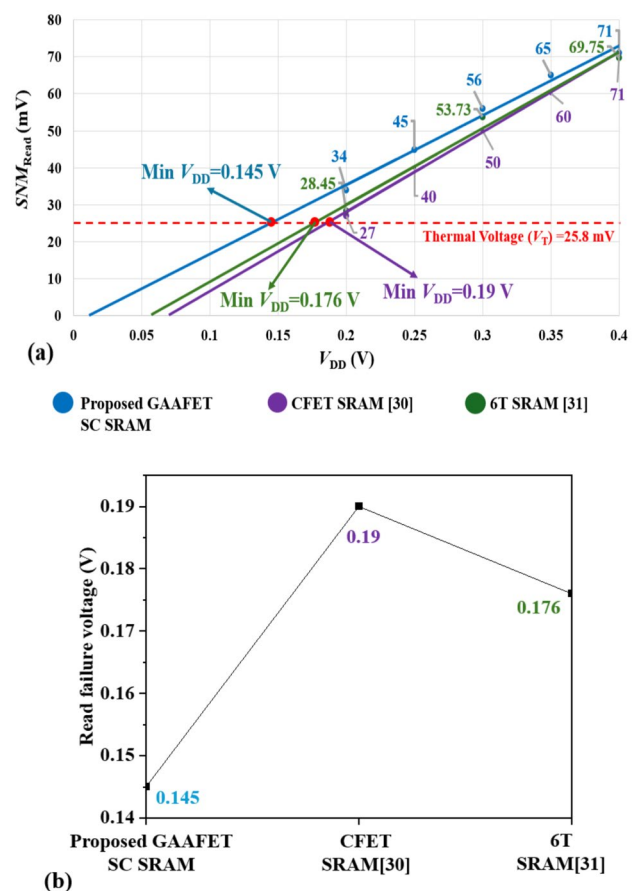
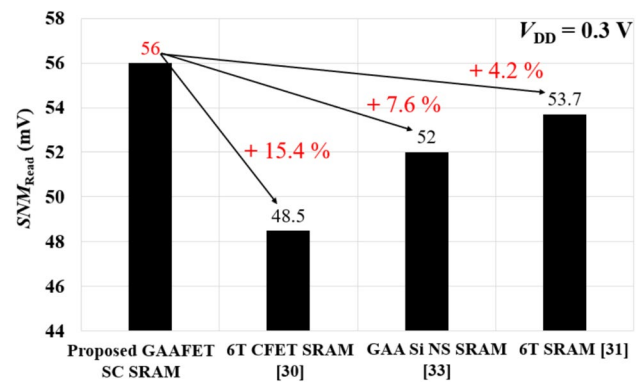


Fig. 24 SNM_{Read} comparison for 6T GAAFET SC SRAM with various conventional counterparts



In Fig. 24, we select the best performing GAAFET PTnMOSFET from the proposed PTnMOSFET devices and compare its SNM_{Read} in the SRAM circuit with different conventional counterparts [30, 31, 33]. Under the conditions of having the same number of transistors and V_{DD} , our SNM_{Read} is 15.4%, 7.6% and 4.2% higher than those reported in references, respectively. This indicates that our GAAFET PTnMOSFET has greater potential in enhancing SRAM read capability.

6 Conclusion

This study presents a PTnMOSFET as a viable alternative to pMOSFETs, offering simplified fabrication by eliminating the need for sizing adjustments or stress engineering techniques. By adjusting the source/drain doping concentrations, the proposed PTnMOSFET effectively addresses current mismatch issues between pMOSFETs and nMOSFETs. Simulation results demonstrate that the GAA PTnMOSFET achieves a wide design window, a low average subthreshold swing (SSavg) of 74 mV/dec, and an $I_{ON}(PTnMOSFET)/I_{ON}(nMOSFET)$ ratio of 0.89, validating its potential for application in advanced technology nodes such as 30 nm and 14 nm. The circuit-level evaluation highlights the advantages of integrating the PTnMOSFET into GAA SC circuits and SRAM designs. The GAA SC circuit shows significant improvements over conventional GAA designs ($W_p/W_n = 1$), achieving shorter delay times, higher voltage gain, and more symmetric signal noise margins. Furthermore, the GAA SC SRAM demonstrates a read failure voltage of 0.145 V and a static signal noise margin of 56 mV, suggesting enhanced write-ability and stability compared to recent stacked nanosheet and cFET-based SRAM designs. While these findings underscore the strengths of the proposed PTnMOSFET architecture, it is important to recognize the limitations of this work. The results presented are based on simulations, and further experimental validation under diverse operating conditions including process, voltage, and temperature variations is necessary to comprehensively assess the robustness and practical applicability of the PTnMOSFET for next-generation semiconductor technologies. Future work should focus on these aspects to establish a broader understanding of the design's scalability and reliability.

Author contributions J.T.L. and Y.Y.C. devised the idea for the study. J.T.L. and Y.Y.C. designed and drafted the figures. J.T.L. and Y.Y.C. prepared the manuscript. All authors reviewed the manuscript.

Data availability The data and analyses generated in this study have been agreed to by all authors and are available from the corresponding authors upon reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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